Implementation of CDM based PID Controller for Magnetic Ball Levitation System

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***Abstract* - In this paper, a new Proportional integral and Derivative (PID) controller is designed based on Coefficient Diagram Method (CDM) for the magnetic ball levitation system. The magnetic levitation system is second order unstable system. Though PID controllers produce a controlled output for unstable systems, the performance of the system in terms of time domain specifications such as overshoot, settling time, Integral Square Error (ISE) and Integral Absolute Error (IAE) are poor. To solve this, CDM based PID controller is proposed. The performance of the proposed CDM-PID controller is analyzed by comparing with the conventional PID controller. The simulation results show that the proposed CDM-PID controller gives better closed loop performance than the conventional PID controller.**

***Keywords – CDM-PID controller, second order, unstable, magnetic ball levitation system*.**

I. INTRODUCTION

A proportional-Integral-Derivative (PID) controller has been widely used in most of the process industries due to their simplicity and robustness [1]. Proper tuning of the PID controller is required to achieve the desired performance of the process. There are various tuning methods available to optimize the PID controller parameters [2]. Most of the tuning rules are related to stable, unstable and integrating process, whereas only a few literatures is available for double integrating systems. Therefore, a new robust controller is required for second order unstable system.

The Co-efficient Diagram Method (CDM) is an algebraic approach that combines classical and modern control theories [3]. This method is developed based on the relationship between the obtained closed loop system characteristics and the placement of poles of its characteristics polynomial on the complex plane. CDM based controller provides a convenient parameter selection rules to design a controller under the conditions of stability, robustness and time domain performances.

Magnetic levitation refers to the methodology in which steel ball is supported by magnetic forces. Due to contactless feature of this technology, it is possible to build high speed devices. Maglev technology has applications in frictionless bearings [4], high speed ground transportation system [5] and vibration isolation devices [6]. In recent days, PID controller is designed for magnetic levitation system based on trial and error method [7]. PID parameters are optimized using PSO algorithms for better performance [8]. The model predictive controller is used for wide working range of nonlinear system [9].

Magnetic levitation system is challenging to control because of its non-linear and their open loop unstable nature. The PID controller cannot provide perfect control for highly nonlinear system. The PID controller requires manual retuning for increasing the performance. Hence the CDM based PID controller is proposed to magnetic levitation system to maintain the ball at the desired position.

II. MAGNETIC BALL LEVITATION SYSTEM

In the magnetic levitation system, the ball is suspended in air by magnetic forces. The magnetic force produced by the electromagnet is opposite to gravity and it maintains the suspended steel ball in a levitated position as shown in figure 1. The magnetic force depends on the electromagnet current and the air gap between the steel ball and the electromagnet.

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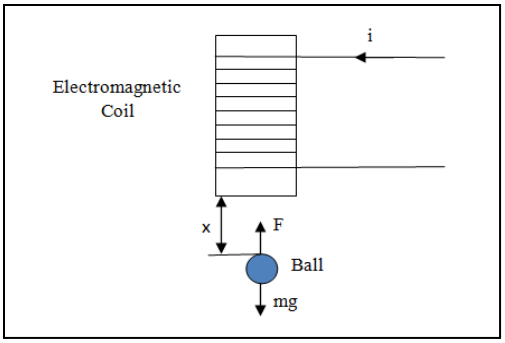

Fig. 1. Schematic illustration of an Electromagnetic Ball Levitation system

The motion of the steel ball in the magnetic field is expressed as [10]

$$m\ddot{x} = mg - c\frac{i^2}{x^2} \quad (1)$$

Where m is the mass of the suspended steel ball, *x* is the air gap between the steel ball and the electromagnet, g is the acceleration due to gravity, c is the magnetic constant and i is the current flowing through the electromagnet coil.

Table I summarizes the variables and parameters used in this problem [11]. Here the equilibrium position of the iron ball is 0.95 – 1.25 cm.

TABLE I. Parameters of Magnetic Ball Levitation system

Parameters	Description	Value
m	Mass of the ball (kg)	0.533
x_0	Nominal air gap (cm)	0.95
i_0	Equilibrium current (A)	1.28
g	Gravitational acceleration (m/s^2)	9.8
c	Magnetic constant (Nm^2/A^2)	$37.75*10^{-5}$

Hence, the transfer function of the nominal plant based on the parameters in table I leads to,

$$G_p(s) = \frac{X(s)}{I(s)} = \frac{-2.009}{s^2-2.7068} \quad (2)$$

The above transfer function shows that the magnetic ball levitation system is unstable.

III. DESIGN OF CONTROLLERS

A. Conventional PID Controller

The performance of conventional PID controller is used as the base for analyzing the performance of proposed controller. The parameters of conventional PID controller such as proportional gain (K_p), integral gain (K_i) and derivative gain (K_d) are computed based on the tuning method proposed by Huang and Chen [12] and is given in Table II.

TABLE II. Conventional PID controller parameters

Tuning Methods	K_p	K_i	K_d
Huang and Chen (1999)	-6.0892	-17.4045	-3.690

B. CDM-PID Controller

The parameters of CDM based PID controller are computed based on design procedure given in (Meenakshipriya et al., 2011) [13]. The structure of CDM based PID controller is shown in figure 2. where, r - input signal, y - output signal, u - control signal, d - external disturbance signal, A(s) - forward denominator polynomial, B(s) - feedback numerator polynomial, F(s) - reference numerator polynomial, $G_p(s)$ - Plant transfer function, N(S) and D(s) are the numerator and denominator polynomials of the plant transfer function, $G_c(s)$ - main controller, $C_f(s)$ - feed-forward controller

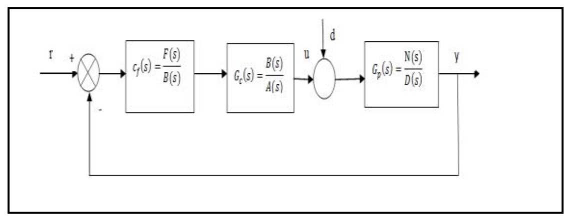

Fig.2. Proposed CDM-PID controller for Magnetic Ball Levitation system

The transfer function of the magnetic levitation system is given in equation (3).

$$G_p(s) = \frac{-2.009}{s^2-2.7068} = \frac{N(s)}{D(s)} \quad (3)$$

For second order system, the controller polynomials A(s) and B(s) are given in equations (4) and (5).

$$A(s) = I_1 S + I_0 \quad (4)$$

$$B(s) = k_2 s^2 + k_1 s + k_0 \quad (5)$$

Here the step disturbance signal is considered. Thus the transfer function of the controller with I_0=0, is given in equation (6),

$$G_c(s) = \frac{B(s)}{A(s)} = \frac{k_2 s^2 + k_1 s + k_0}{I_1 S + I_0} \quad (6)$$

where, k_2,k_1,k_0,I_1 and I_0 are the controller design parameters.

The characteristic polynomial of the closed loop system is defined as

$$P(s) = A(s)D(s) + B(s)N(s)$$

$$P(s) = (I_1 S)(s^2 - 2.7068) + (k_2 s^2 + k_1 s + k_0)(-2.009)$$

$$P(s) = I_1 s^3 - 2.009 k_2 s^2 - 2.009 k_1 s - 2.7068 I_1 s - 2.009 k_0 \quad (7)$$

In CDM, the target characteristic polynomial is given as,

$$p_{target}(s) = \frac{\tau^3}{\Upsilon_2 \Upsilon_1^2} s^3 + \frac{\tau^2}{\Upsilon_1} s^2 + \tau s + 1 \quad (8)$$

where, τ - time constant and Υ_i - stability indices.

The speed of the closed loop system depends on equivalent time constant τ. According to the Manabe (1998), the design parameters are given in equations (9) and (10).

$$\tau = t_s / (2.5 \approx 3) \quad (9)$$

where t_s is the user defined settling time. The settling time is considered as 2 sec.

$$\Upsilon_i = [2.5\ 2\ \ldots.] \text{ where } I = 1,2,\ldots n-1,\ \Upsilon_0 = \Upsilon_\infty = 0 \quad (10)$$

To achieve a good closed loop response, the polynomial F(s) is chosen as follows,

$$F(s) = \frac{p(s)}{N(s)} \Big/ s = 0 = \frac{-2.009 k_0}{-2.009} = k_0 \quad (11)$$

By equating the coefficients of the terms of the equal power of equation (7) and (8), the CDM controller parameters I_1, K_0, K_1 and K_2 are computed as follows

$$I_1 = \frac{\tau^3}{\Upsilon_2 \Upsilon_1^2}; \quad k_0 = \frac{-1}{2.009} = -0.497; \quad k_1 = \frac{\tau + 2.7068 I_1}{-2.009}; \quad k_2 = \frac{-\tau^2}{\Upsilon_1 * 2.009} \quad (12)$$

B.1. CDM based PID controller parameters

For conventional PID controller, the main controller $G_c(s)$ is chosen as,

$$G_c(s) = K_c\left(1 + \frac{1}{T_i s} + T_d s\right) \quad (13)$$

Substitute the equation (13) in (6),

$$K_c\left(1 + \frac{1}{T_i s} + T_d s\right) = \frac{k_2 s^2 + k_1 s + k_0}{I_1 s} \quad (14)$$

By equating the coefficients of like power terms, the PID controller parameters based on CDM are obtained as follows,

$$K_c = \frac{K_1}{I_1};\ T_i = \frac{k_1}{k_0};\ T_d = \frac{k_2}{k_1} \quad (15)$$

By using the equations (5) and (11), the feed-forward controller is found to be

$$c_f(s) = \frac{F(s)}{B(s)} = \frac{k_0}{k_2 s^2 + k_1 s + k_0} \quad (16)$$

B.2. Tuning of Time constant τ and stability indices

The selection of time constant τ and stability indices Υ_1 and Υ_2 are playing a predominant role to obtain the optimal controller parameters for CDM-PID controller. The effect of time constant is analyzed through simulation for the range of τ from 0.5 to 10 at the operating point of ball position as 1 cm. If we further decrease the value of τ, it will increase the settling time. According to the Manabe's recommendation, the stability indices are chosen as Υ_1 = 2.5 and Υ_2 = 2. The closed loop responses are recorded as shown in figure 3 by increasing the value of time constant τ at the sampling interval of 1. From the recorded response, the performance measures are calculated and are tabulated in Table III . From the table III, it is clear that at τ=0.5, the CDM - PID controller provides good error indices as well as quality indices compared to other τ values.

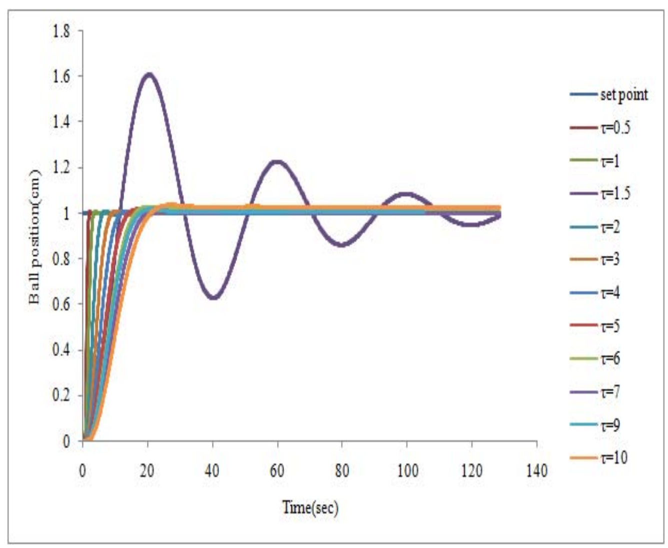

Fig. 3. Effect of time constant τ for stability indices Υ_1 = 2.5 and Υ_2 = 2

TABLE III. Performance measures for the effect of time constant τ for stability indices $\Upsilon_1 = 2.5$ and $\Upsilon_2 = 2$

Time constant	Performance measures			
	Peak overshoot (%)	*Settling time(sec)*	*ISE*	*IAE*
τ= 0.5	0.7	6.72	17.43	20.22
τ=1	0.8	11.76	25.38	30.91
τ=1.5	60	--	--	--
τ=2	0.85	16.92	41.30	52.47
τ=3	0.9	12.84	57.34	74.30
τ=4	0.83	16.5	73.52	96.34
τ=5	1.6	116.88	89.83	124.73
τ =6	2.7	116.9	106.23	159.66
τ =7	1.3	85.32	122.61	167.36
τ =9	1.8	104.94	109.09	154.70
τ =10	3.6	---	--	---

By setting the time constant τ=0.5 and $\Upsilon_2 = 2$, the effect of Υ_1 on closed loop response at the operating point of 1 cm (Desired Ball Position) is recorded as shown in figure 4 and the performance measures are tabulated in TABLE IV . From the table IV, it is clear that, the CDM-PID controller provides fair response at τ=0.5 and Υ_1= 2.5.

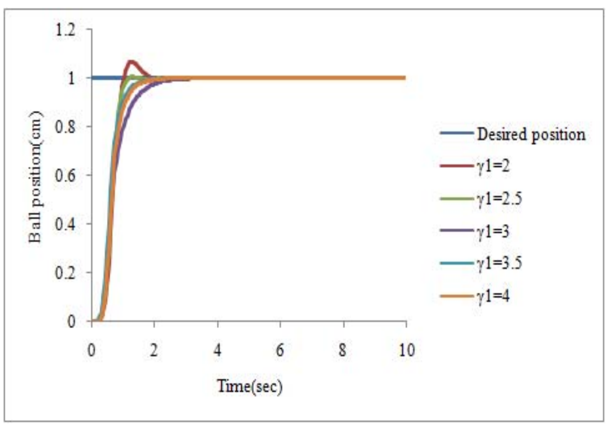

Fig.4. Effect of stability index Υ_1 for τ=0.5 and $\Upsilon_2 = 2$

TABLE IV. Performance Measures for the Effect of stability index Υ_1 for τ=0.5 and $\Upsilon_2 = 2$

Stability Index Υ_1	Performance measures			
	Peak overshoot (%)	*Settling time(sec)*	*ISE*	*IAE*
γ1= 2	6.8	2.82	9.87	12.10
γ1=2.5	Nil	2.76	9.31	11.08
γ1=3	Nil	4.44	9.58	13.15
γ1=3.5	Nil	3.72	8.69	11.09
γ1=4	Nil	3.24	9.49	12.01

To optimize the value of Υ_2, the same procedure is repeated with τ=0.5 and Υ_1= 2.5. The results are recorded as shown in figure 5 and the performance measures are tabulated in Table V.

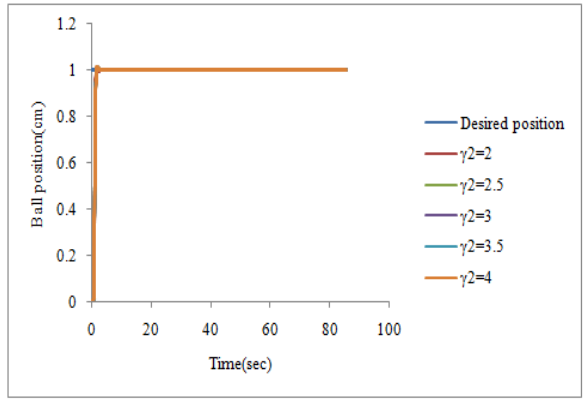

Fig.5. Effect of stability index Υ_2 for τ=0.5 and Υ_1= 2.5

TABLE V. Performance measures for the effect of stability index Υ_2 for τ=0.5 and Υ_1= 2.5

Stability Index Υ_2	Performance measures			
	Peak overshoot (%)	*Settling time(sec)*	*ISE*	*IAE*
γ2= 2	0.8	3.06	15.11	16.86
γ2=2.5	0.65	2.16	15.02	16.81
γ2=3	0.79	3.42	14.98	16.90
γ2=3.5	0.9	3.54	14.93	16.92
γ2=4	0.93	3.72	14.88	16.95

By comparing the above results, it is concluded that the proposed CDM-PID controller provides good performance with τ=0.5, Υ_1= 2.5 and Υ_2 = 2.5. By using these optimized values, the CDM-PID controller parameters for magnetic ball levitation system are calculated and tabulated in Table VI.

TABLE VI. CDM-PID controller parameters

Parameters	Values
Proportional gain K_p	-32.375
Integral gain K_i	-62.14
Derivative gain K_d	-6.183

The transfer function of feed-forward controller $C_f(s)$ is given as,

$$c_f(s) = \frac{-0.497}{-0.497s^2 - 0.259s - 0.497} \qquad (17)$$

IV. SIMULATION RESULTS

The simulation is carried out to analyse the performance of CDM-PID controller with conventional PID controller. The performance of CDM-PID controller and conventional PID controller at the operating point of 1 cm (desired position) are recorded in figure 6 and the performance measures are tabulated in Table VII. From the table VII, it is clear that CDM-PID controller forces the set point at short duration of time and maintains the steady state with minimum overshoot as compared to conventional PID controller.

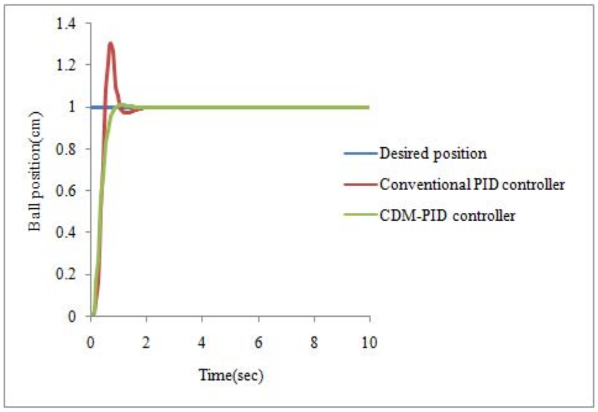

Fig. 6. Performance Conventional PID and CDM-PID controller at the operating point of ball position as 1cm

TABLE VII. Performance measures of Conventional PID and CDM-PID controller at the operating point of ball position as 1cm

Controllers	**Performance measures**			
	Peak overshoot (%)	***Settling time(sec)***	***ISE***	***IAE***
Conventional PID	30	2.22	5.62	7.99
CDM-PID	0.8	0.96	4.90	6.62

V. CONCLUSIONS

In this work, CDM based PID controller is implemented for Magnetic Ball Levitation System in MATLAB SIMULINK platform and compared with conventional PID controller. From the simulation results, it is observed that the perfect control on the desired position of levitated ball supported by magnetic force is made by CDM based PID controller. The proposed CDM-PID controller provides fair transient and steady state response with minimum error indices (ISE and IAE) and good quality indices (peak overshoot and settling time) than the conventional PID controller.

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